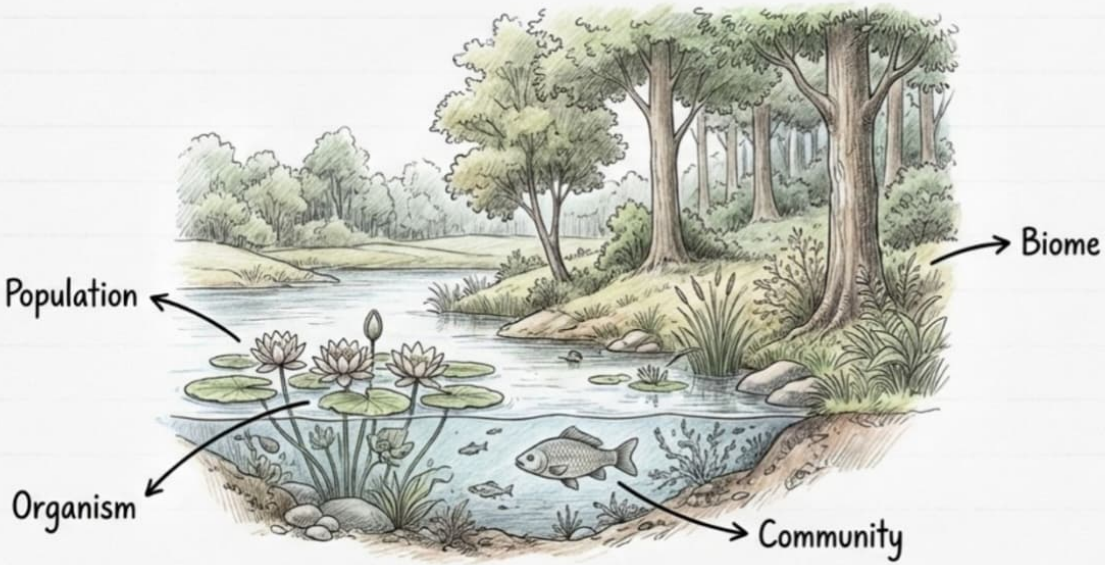
ORGANISMS AND POPULATIONS: ECOLOGY



Definition: Ecology

The study of interactions among organisms and between the organism and its physical (abiotic) environment.

KEY POINTS (NCERT)

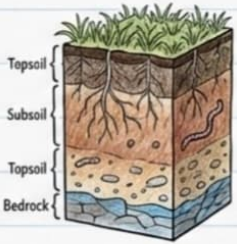
- Ecology operates at four levels of biological organisation: Organisms, Populations, Communities, and Biomes.
- Ramdeo Misra is revered as the Father of Ecology in India, laying the foundations for understanding tropical communities

MAJOR ABIOTIC FACTORS

Light: Photoperiod cues for foraging & migration.

Temperature: Affects enzyme kinetics & basal metabolism.

Water: Salinity: <5 inland, 30-35 sea, >100 hypersaline



Soil: Topography dictates benthic zones.

Key Terms:

Eurythermal / Euryhaline (wide tolerance)

vs.

Stenothermal / Stenohaline (narrow tolerance)

KEY POINTS (NCERT)

- The physico-chemical (abiotic) components alone do not completely characterize a habitat; biotic interactions are also constant.
- Life on earth originated in water; for aquatic organisms, the chemical composition and pH are critical for survival.

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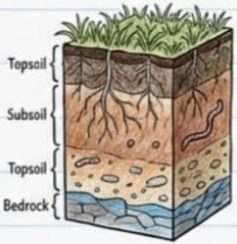

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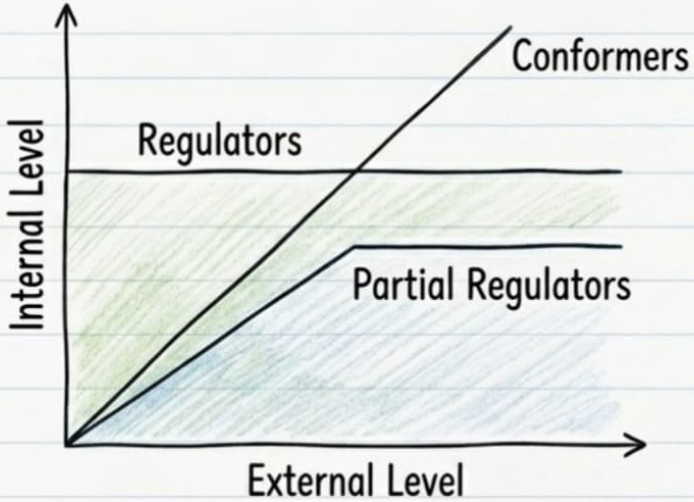
RESPONSES TO ABIOTIC STRESS



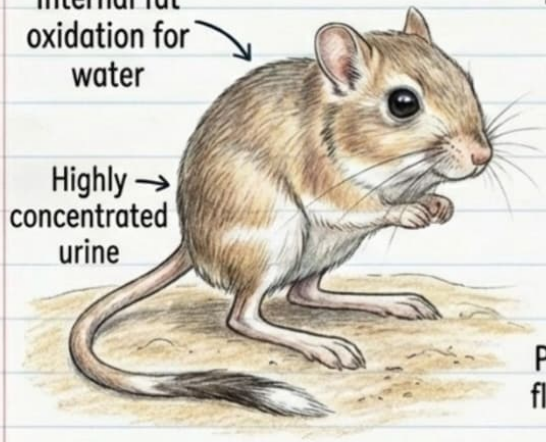
Definition: Homeostasis

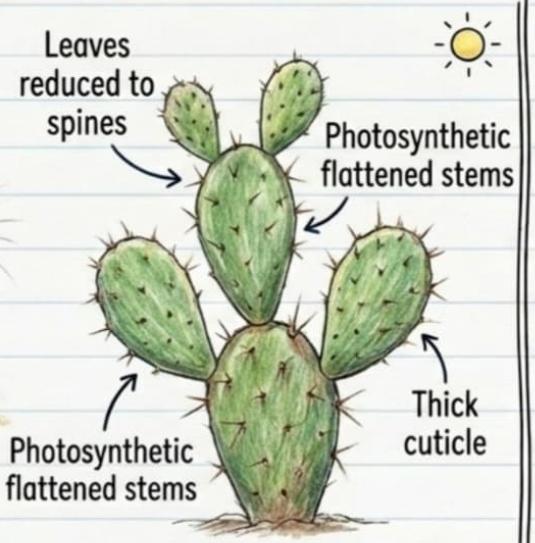
Maintaining a constant internal environment despite varying external conditions.
(e.g., Humans maintain 37°C).

KEY POINTS (NCERT)

- Regulate: Maintain constant temperature/osmolarity (e.g., mammals, birds).
- Conform: Internal environment changes with ambient conditions.
- Suspend: Escape in time via a Hibernation (winter), Aestivation (summer)

ADAPTATIONS IN NATURE





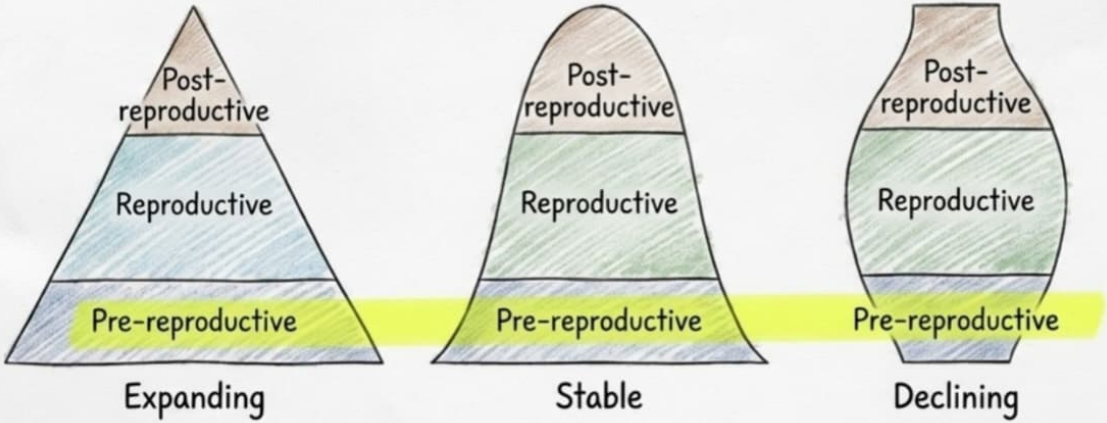
Allen's Rule:

Mammals from colder climates have shorter ears and limbs to minimize heat loss.

KEY POINTS (NCERT)

- Adaptations can be morphological, physiological, or behavioural.
- Altitude sickness is countered physiologically by increasing red blood cell (RBC) production.
- Archaebacteria possess adaptations allowing them to flourish in hot springs exceeding 100°C.

AGE STRUCTURE & PYRAMIDS

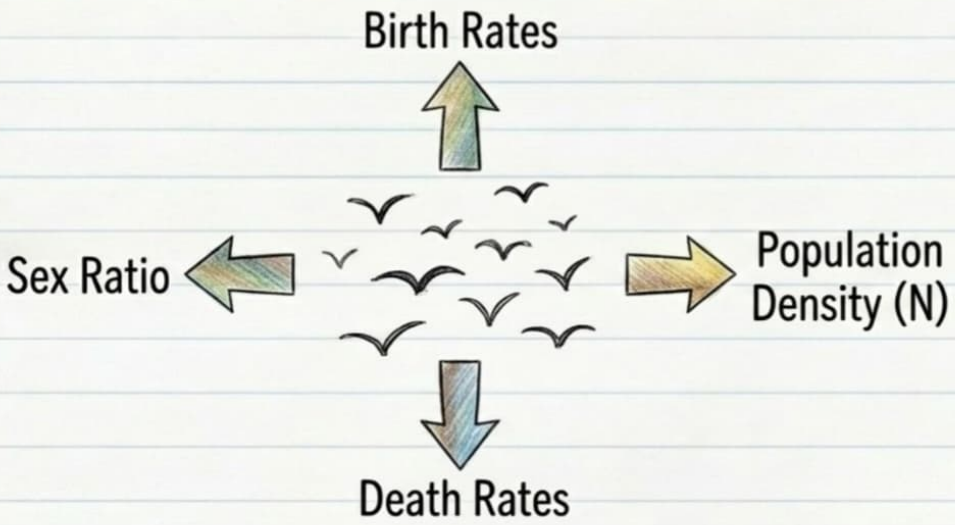


Insight:
The shape of the pyramids reflects the growth status of a population.

KEY POINTS (NCERT)

- Populations are composed of individuals of different ages.
- Plotting the age distribution for a population yields an age pyramid.
- These pyramids are vital for predicting whether a population will grow, stabilize, or decline in the future.

UNDERSTANDING POPULATIONS

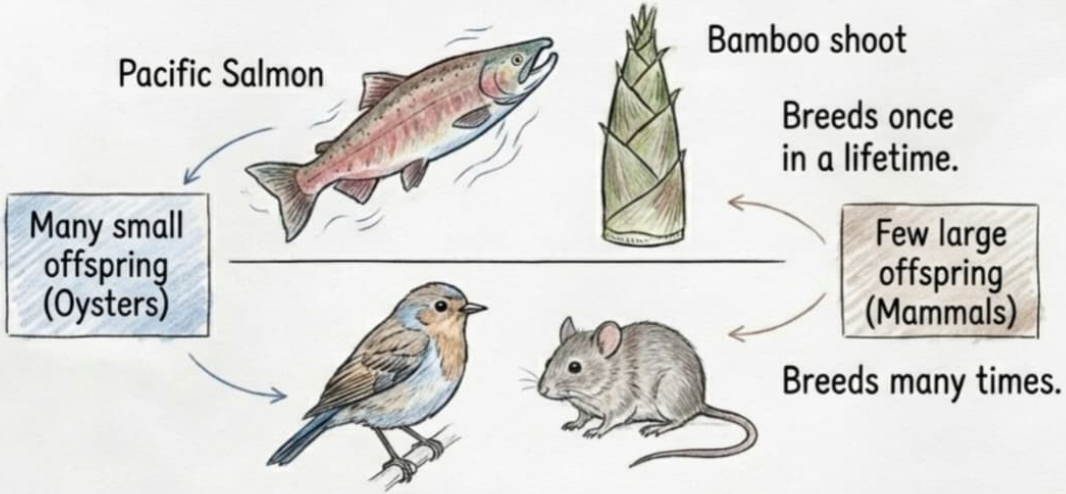


Formula: Rates are expressed as per capita changes.
Example: If 20 lotus plants add 8 new plants, the birth rate is 8/20 = 0.4 offspring per lotus per year.

KEY POINTS (NCERT)

- Populations share/compete for resources and potentially interbreed.
- Individuals have births and deaths; populations have birth rates and death rates.
- Density (N) isn't always counted directly; indirect methods include tiger pug marks or fecal pellets

LIFE HISTORY VARIATION



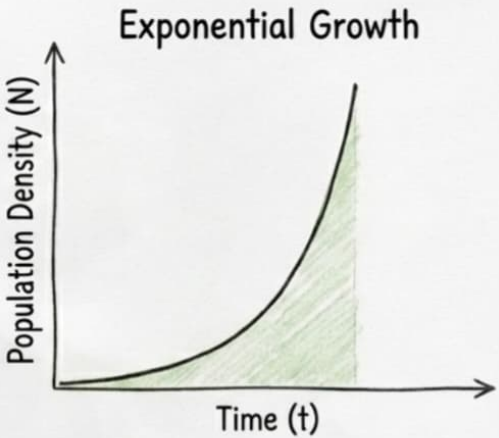
Definition:

Darwinian Fitness
Maximizing reproductive success (high 'r' value) under specific environmental selection pressures.

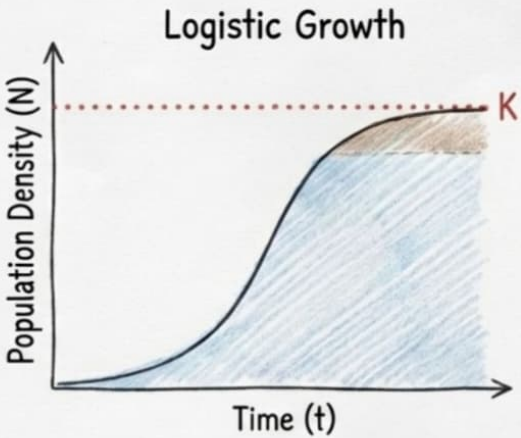
KEY POINTS (NCERT)

- Populations evolve toward the most efficient reproductive strategy for their specific habitat.
- Traits evolve in relation to the constraints imposed by both the abiotic and biotic components of the environment.

POPULATION GROWTH CURVES



Resources are unlimited.



Resources become limiting.

Equations:

Exponential:

$$\frac{dN}{dt} = rN$$

Logistic (Verhulst-Pearl):

$$\frac{dN}{dt} = rN\left[\frac{(K-N)}{K}\right]$$

KEY POINTS (NCERT)

- r = Intrinsic rate of natural increase (e.g., Norway rat = 0.015, Flour beetle = 0.12).
- Exponential growth occurs when resources are unlimited.
- Logistic growth is more realistic; finite resources limit growth at Nature's Carrying Capacity (K)

INTERSPECIFIC INTERACTIONS

Species A	Species B	Name of Interaction
(+)	(+)	Mutualism
(-)	(-)	Competition
(+)	(-)	Predation / Parasitism
(+)	(0)	Commensalism
(-)	(0)	Amensalism

Definitions:

(+) Beneficial

(-) Detrimental

(0) Neutral

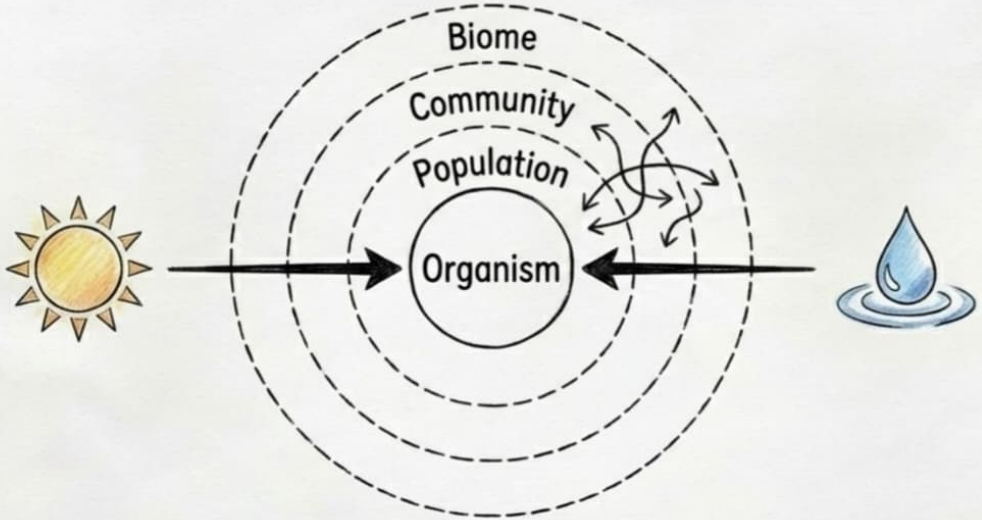
KEY POINTS (NCERT)

- Competition significantly lowers the fitness ('r') of both interacting species.
- Predation acts as a critical conduit for energy transfer across trophic levels.
- Parasites often tightly co-evolve with their specific host species.

Biotechnology: Principles and Processes

Process Chronology & Master Notes

THE ECOLOGICAL WEB (SUMMARY)



Final Thought:

Natural selection operates at the population level, making population ecology the vital bridge to evolution.

KEY POINTS (NCERT)

- Ecology spans from individual physiological adaptations to complex community interactions.
- Populations are highly dynamic, governed by births, deaths, immigration, emigration, and carrying capacity (K).
- Ultimate survival depends on successfully managing abiotic stress and navigating complex interspecific biotic